

UBT310: BIOCHEMISTRY-II

L	T	P	Cr
3	0	0	3.0

Course Objective: To make Students understand interaction of biomolecules in cell, biotransformations and enzyme-catalyzed metabolic pathways obeying physical and chemical laws to maintain and perpetuate life forms.

Syllabus

Bioenergetics: Basic principles of thermodynamics, Common biochemical reactions, Concepts of energy and energy change in biochemical processes, Factors affecting free energy changes in biochemical reactions, Group transfer potential, Role of ATP as energy currency, High energy phosphates and thioesters in biological systems, Biological oxidation-reduction reactions

Enzymes: Nomenclature of enzymes, Enzyme kinetics, Mechanism of enzymatic, Catalysis, Active site, Activators and inhibitors, Coenzymes, Isoenzymes, Michaelis-Menten equation, K_m and V_{max} value, Regulation of enzyme activity (single-substrate and multi-substrate reactions)

Intermediary Metabolism: Basic concept and design of metabolism, Regulation of metabolic pathways, Basic carbohydrate metabolism: glycolysis, TCA cycle, pentose phosphate pathway, gluconeogenesis, and glycogen metabolism; Electron transport and oxidative phosphorylation; Photosynthesis; Fatty acid and lipid metabolism; Metabolism of amino acids, purines, pyrimidines and nucleotides

Biosignaling: General features of signal transduction, G protein-coupled receptors, and second messengers, Receptor Tyrosine kinases, Gated ion channels, signaling by steroid hormones, Cell cycle, Regulation of cell cycle by protein kinases, programmed cell death

Integration of Metabolic Pathways: Hormonal control, inter-relationships between carbohydrate, protein, lipid and nucleic acid metabolism

Course Learning Objectives (CLO)

The students will be able to:

1. Know the various signals that influence different cellular/metabolic processes.
2. Realize that all the cellular/biochemical changes obey the basic thermodynamic principles.
3. Explain release of free energy during catabolic breakdown of the substances and its utilization during anabolic pathways.
4. Comprehend role of hormones in the integration of metabolic pathways.
5. Perform various experiments related to biochemistry.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

6. Comprehend role of enzymes as biocatalysts and mechanisms of enzyme catalysis.

Text Books

1. Nelson, D.L. and Cox, M.M., Lehninger Principles of Biochemistry, W.H. Freeman (2008) 5th ed.
2. Jain, J.L., Jain, S. and Jain, N., Fundamentals of Biochemistry, S. Chand and Company Ltd. (2005).
3. Rao, B.S. and Deshpande, V., Experimental Biochemistry: A Students companion. Anshan Publication (2005).
4. Wilson, K. and Walker, J., Practical Biochemistry, Principles and Techniques, Cambridge University Press (1995) 5th ed.

Reference Books

1. Berg, J.M., Tymoczko, J.L. and Stryer, L., Biochemistry, W. H. Freeman (2006) 6th ed.
2. Campbell, M.K. and Farrell, S.O., Biochemistry, Brooks Cole (2006) 5th ed.
3. Switzer, R.L. and Garritty, L.F., Experimental Biochemistry, W. H. Freeman (1999) 3rd ed.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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